

Theoretical Foundations of Reinforcement Learning:

Week 3

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Dynamic Programming Algorithms

Policy Evaluation

Given a fixed policy π , compute $V_\pi(s)$ for all states by repeatedly applying the Bellman expectation equation until convergence.

Algorithm 1 Policy Evaluation

```
1: Initialise  $V(s) = 0$  for all  $s \in S$ 
2: repeat
3:    $\Delta \leftarrow 0$ 
4:   for each  $s \in S$  do
5:      $v \leftarrow V(s)$ 
6:      $V(s) \leftarrow \sum_a \pi(a|s) \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V(s')]$ 
7:      $\Delta \leftarrow \max(\Delta, |v - V(s)|)$ 
8:   end for
9: until  $\Delta < \theta$   $\triangleright \theta$  is a small convergence threshold
10: return  $V$ 
```

Policy Iteration

Alternate between evaluating the current policy and improving it greedily until the policy stops changing.

Value Iteration

Combine evaluation and improvement into a single update. No need to evaluate the policy fully before improving.

The only difference from policy evaluation is $\max_a$ instead of $\sum_a \pi(a|s)$ — value iteration always picks the best action at each step.

Algorithm 2 Policy Iteration

```
1: Initialise  $\pi(s)$  arbitrarily for all  $s \in S$ 
2: repeat
3:    $V \leftarrow \text{POLICYEVALUATION}(\pi)$ 
4:    $\text{policy-stable} \leftarrow \text{true}$ 
5:   for each  $s \in S$  do
6:      $a_{\text{old}} \leftarrow \pi(s)$ 
7:      $\pi(s) \leftarrow \arg \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V(s')]$ 
8:     if  $a_{\text{old}} \neq \pi(s)$  then
9:        $\text{policy-stable} \leftarrow \text{false}$ 
10:    end if
11:  end for
12: until  $\text{policy-stable}$ 
13: return  $\pi$ 
```

Algorithm 3 Value Iteration

```
1: Initialise  $V(s) = 0$  for all  $s \in S$ 
2: repeat
3:    $\Delta \leftarrow 0$ 
4:   for each  $s \in S$  do
5:      $v \leftarrow V(s)$ 
6:      $V(s) \leftarrow \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V(s')]$ 
7:      $\Delta \leftarrow \max(\Delta, |v - V(s)|)$ 
8:   end for
9: until  $\Delta < \theta$ 
10: Extract policy:  $\pi(s) \leftarrow \arg \max_a \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma V(s')]$ 
11: return  $\pi$ 
```

MC vs TD on a Toy MDP

MDP Setup

Consider a simple 3-state MDP: $S = \{s_1, s_2, s_3\}$ where s_3 is the terminal state.

- From s_1 : always transition to s_2 , reward $R = 0$
- From s_2 : always transition to s_3 , reward $R = 1$
- s_3 : terminal, $R = 0$

Set $\gamma = 1$ (no discounting) and $\alpha = 0.5$ (learning rate). Initialise $V(s_1) = V(s_2) = 0$.

The true values are: $V(s_1) = 1$, $V(s_2) = 1$.

Single Episode

The only possible episode is: $s_1 \xrightarrow{R=0} s_2 \xrightarrow{R=1} s_3$

Monte Carlo Update

MC waits for the episode to end, computes actual returns, then updates.

Return from s_1 : $G_1 = 0 + 1 = 1$

Return from s_2 : $G_2 = 1$

Updates:

$$V(s_1) \leftarrow 0 + 0.5[1 - 0] = 0.5$$

$$V(s_2) \leftarrow 0 + 0.5[1 - 0] = 0.5$$

Both states move toward the correct value of 1 in one episode, but only halfway due to $\alpha = 0.5$.

TD(0) Update

TD updates immediately after each step using the next state's current estimate.

Step 1: transition $s_1 \rightarrow s_2$, reward $R = 0$

$$V(s_1) \leftarrow 0 + 0.5[0 + V(s_2) - 0] = 0 + 0.5[0 + 0 - 0] = 0$$

$V(s_1)$ does not change because $V(s_2) = 0$ at this point.

Step 2: transition $s_2 \rightarrow s_3$, reward $R = 1$

$$V(s_2) \leftarrow 0 + 0.5[1 + V(s_3) - 0] = 0 + 0.5[1 + 0] = 0.5$$

After one episode: $V(s_1) = 0$, $V(s_2) = 0.5$.

Comparison

After one episode, MC correctly updates both $V(s_1)$ and $V(s_2)$ to 0.5. TD only updates $V(s_2)$ to 0.5 and leaves $V(s_1) = 0$, because when s_1 was visited, $V(s_2)$ was still 0. TD requires multiple episodes for value estimates to propagate backwards through the state chain, whereas MC propagates the full return in a single episode.

This illustrates the bias in early TD estimates: $V(s_2) = 0$ was wrong, and TD used it to update $V(s_1)$, causing no update at all. MC used the true return and was correct immediately.